

Dear Reader,

Thank you for reading my paper, it truly means a lot. I hope my research can help you in any way, whether it's lessening your EV qualms or helping you decide on your next vehicle purchase, or maybe make it easier for you to judge tesla drivers on the road.

As of right now, I haven't written anything for the paper yet, so any ideas you have or mistakes you think need correcting, go for it. I'm not sure what I am looking for yet since I haven't made any progress, but I think I am looking for any ideas, and suggestions regarding the structure and readability of this paper.

I am focused on the integrity of my sources and thinking of the most possible ways to help the target audience. This paper is focused on solutions for EV range issues. My experience with my car has inspired me to focus on this topic because the range I have gave me a lot of frustrations. I hope to find credible sources and reliable ways to solve my issue and improve my range, and I hope it will help someone else who might need it too.

Understanding and Solving Low Mileage Range in Electric Vehicles

Tianyi Zhang

Electric vehicles have been marketed to the public with the promises that there will be lower operating prices (no gas), lower repair costs, and a cleaner environment. For many buyers, the advertised range on the window sticker is an important part of the purchase. A Tesla Model 3 Long Range, for example, is rated by the EPA at 358 miles per charge, more than sufficient for daily driving. But for many owners, including myself, the reality hits differently. On cold winter days, or after a long highway stretch, the battery percentage drops far faster than expected, and the advertised number begins to feel like a huge disappointing lie. This experience is unfortunately very common. Range anxiety is one of the most frequently cited issues with new EV ownership, and there is often a gap between rated and real world range. Range anxiety exists because the advertised range shoppers see is not always what they turn out to be. Fortunately, these negative factors in daily use can be solved by fixing driving habits so that you can actually get the most out of every charge.

To understand the problem, we need to understand where the advertised numbers come from. In the United States, the EPA tests vehicle range using a standardized laboratory procedure that simulates a combination of city and highway driving under controlled temperature conditions, typically around 75 degrees Fahrenheit. These tests are designed to provide a consistent baseline for comparing vehicles, not to perfectly replicate real-world conditions (U.S. Department of Energy, 2023). The result is that the EPA figure represents a range based on the best possible driving scenarios, one that does not represent daily commutes for most drivers.

This discrepancy is well documented. In a 2021 study, Recurrent Auto analyzed real world range data from over 7,000 Tesla owners across different models and found that actual range routinely fell 10 to 30 percent below EPA estimates depending on conditions, with models performing significantly worse in winter months (Recurrent Auto, 2021). Although this isn't groundbreaking news to owners and well versed shoppers, it's still a crippling issue. This issue unknowingly creates range anxiety by producing false expectations of range. Tesla acknowledges this in its own documentation, noting in the owner's manual that "driving, charging, and temperature conditions" all influence the displayed range estimate and that the rated range is based on "EPA test results" which "may not reflect your actual driving results" (Tesla, Inc., 2023, p. 164).

There are many factors that affect EV range. Temperature is usually the biggest culprit, affecting the Lithium ion batteries directly. The electron particles in the battery are highly sensitive to cold, and in low temperatures, the chemical reactions inside the battery slow down, reducing the amount of energy it can deliver. Not only that, AC usage also goes up in dramatic temperatures.

Driving habits are the second common fatality for EV range. Aerodynamic drag increases with the square of speed, meaning that highway driving at 80 mph consumes dramatically more energy per mile than cruising at 60 mph. The U.S. Department of Energy notes that driving at 70 mph instead of 60 mph can reduce EV range by roughly 14 percent (U.S. Department of Energy, 2023). Aggressive acceleration, made frankly quite easy with electric motors, also consumes lots of energy. Carrying heavier loads, AC usage, cold temperatures, speeding and quick accelerations all compound together and dramatically reduces the EPA estimated range. An EV driven like a sports car in city traffic will see its range suffer accordingly.

Beyond temperature and speed, a host of smaller factors add up. Running seat heaters, a stereo system, phone charging, and cabin lights all pull from the same battery pack that moves the car. Tesla's own data shows that heating the cabin from cold can draw 3 to 5 kilowatts continuously, a significant load on a battery pack that can only deliver around 20 to 25 kW in normal city driving. Additionally, consistently charging the battery to 100 percent or discharging it nearly to zero produces long term battery degradation, which permanently reduces the usable capacity of the pack over time (Tesla, Inc., 2023).

One of the most effective and underused strategies is preconditioning using the car's climate control system while it is still plugged in, so the cabin reaches a comfortable temperature before the driver unplugs. Tesla's manual specifically recommends preconditioning "before leaving" in cold weather to maximize range (Tesla, Inc., 2023, p. 165).

Another way to extend range is to cruise slower on the highway. Driving at 65 mph instead of 80 mph on a long trip can add 30 to 50 miles of range on a modern Tesla, although this strategy is not ideal and often ignored by most drivers. Tesla's Autopilot and its trip planner are designed to optimize efficiency however, so if you do decide to wait a little longer before reaching your destination, the car will navigate the best route and drive itself there for you.

Rolling resistance is also a meaningful factor in energy consumption. Underinflated tires increase resistance and reduce range, therefore, keeping all 4 tires inflated to the optimal PSI, 42 (when cold), will save your range a noticeable amount. The U.S. Department of Energy estimates that for every 1 PSI drop in tire pressure across all four tires, fuel economy (or in EV terms, range efficiency) drops by approximately 0.2 percent. This small number adds up over many miles and many months of driving (U.S. Department of Energy, 2023).

Another way to increase aerodynamics and decrease wind resistance is to use the aerodynamic wheel covers, the flat rims you see on some teslas. These rims aren't just ugly for no reason, but they actually help the air pass around your wheels rather than going into them and restricting airflow.

Another common driving habit that can extend range is to use regenerative braking as much as possible. Regenerative braking converts the kinetic energy of deceleration back into electrical energy, partially recharging the battery every time the driver lifts off the accelerator and cruises to slow down, rather than a harsh stop. Some in car settings that promote range health is Tesla's "Standard" regenerative braking setting. Paired with the "Chill" drive mode, which limits maximum acceleration and limits peak power draw. These settings are accessible through the car's touchscreen and require no additional hardware or cost.

The frustration of watching an EV's range fall short of its advertised number is a pain, and it is shared by the growing population of electric vehicle owners, but it is not an unsolvable problem that should deter someone away from leaving ICE vehicles behind and driving electric. Understanding the problem and preparing for the issue is something buyers should do research on before purchasing an EV. The range gap between the EPA sticker and the real world is the product of specific, unfortunate factors of physics. Cold temperatures, high speeds, aerodynamics, and battery degradation over time. Each of these problems have practical countermeasures, preconditioning, slower driving speeds and acceleration, proper tire maintenance, smart charging habits, and regenerative braking, that all EV drivers today should pick up, at no additional cost. The goal of this paper is not to discourage EV ownership, but quite the opposite. Understanding why range drops and how to fight back against it is what transforms a frustrating ownership experience into a confident one. The relationship between EV owners

and their cars should be happy and comfortable. The miles are there, drivers just need to know how to claim them.

Works Cited

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